

The Kernel Theorem

Let $T: V \rightarrow W$ be a linear transform w/ kernel $\text{Ker}(T)$. Then

(i) $\text{Ker}(T)$ is a subspace of V .

(ii) T is injective iff $\text{Ker}(T) = \{\vec{0}\}$

Pf: (i) was done many times, and is also a practice final question.

(ii) If T is injective

$$T(\vec{u}) = T(\vec{v}) \Rightarrow \vec{u} = \vec{v}.$$

so $T(\vec{w}) = \vec{0}$ means $\vec{w} \in \text{Ker}(T)$

$$\text{so } T(\vec{w}) = T(\vec{0}) \Rightarrow \vec{w} = \vec{0}$$

$\therefore$ only $\vec{0} \in \text{Ker}(T)$!

If $\text{Ker}(T) = \{\vec{0}\}$ Suppose $T(\vec{u}) = T(\vec{v})$. Then

$$T(\vec{u}) - T(\vec{v}) = T(\vec{u} - \vec{v}) = \vec{0}$$

This means $\vec{u} - \vec{v} \in \text{Ker}(T)$. But only $\vec{0} \in \text{Ker}(T)$ so $\vec{u} - \vec{v} = \vec{0} \Rightarrow \vec{u} = \vec{v}$ ✓

Ex// Same matrix. $A = \begin{bmatrix} 2 & -4 & 3 & 6 \\ -1 & 2 & -2 & -3 \end{bmatrix}$

The kernel is all $\vec{v}$ st. $A\vec{v} = \vec{0}$. So we want

$$\left[\begin{array}{cccc|c} 2 & -4 & 3 & 6 & 0 \\ -1 & 2 & -2 & -3 & 0 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{cccc|c} 1 & -2 & 0 & 3 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{array} \right] \quad \begin{array}{l} \text{basic var. : } v_1, v_3 \\ \text{free var. : } v_2, v_4 \end{array}$$

$$\Rightarrow v_3 = 0, \quad v_1 = 2v_2 - 3v_4 \quad \text{so} \quad \vec{v}_h = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{pmatrix} = \begin{pmatrix} 2v_2 - 3v_4 \\ v_2 \\ 0 \\ v_4 \end{pmatrix} = v_2 \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + v_4 \begin{pmatrix} -3 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Those two vectors form a basis for the null space.
The nullity, then, is 2.

Ex// $\ddot{x} + \dot{x} - 6x = 0$. Note: $L(x) = 0$
w/ $L(\cdot) = \left[\frac{d^2}{dt^2} + \frac{d}{dt} - 6 \right](\cdot)$

So (as usual) assume $x = e^{\lambda t}$, then

$$\lambda^2 + \lambda - 6 = (\lambda - 3)(\lambda + 2) = 0 \Rightarrow \lambda = \{-2, 3\}$$

$$\therefore x_h = c_1 e^{-2t} + c_2 e^{3t}$$

The vectors $\{e^{-2t}, e^{3t}\}$ form a basis for the kernel of the linear Transform L .

// The Rank-Nullity Theorem, (final form)

Let $T: V \rightarrow W$ be a linear transform. Then

$$\dim(V) = \dim(\text{Im}(T)) + \dim(\text{Ker}(T)) \quad \checkmark \checkmark$$

Compare w/ the previous example. $\dim(\mathbb{R}^4) = 4$

$$\dim(\text{Im}(T)) = 2 \quad ; \quad \dim(\text{Ker}(T)) = 2$$

$$4 = 2 + 2 \quad \checkmark \checkmark$$

Eigenvalues & Eigenvectors

Consider a first-order ODE in 2D given by

$$\frac{du}{dt} = au + bv$$

$$\frac{dv}{dt} = cu + dv \quad \text{w/} \quad \begin{pmatrix} u(0) \\ v(0) \end{pmatrix} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$$

or alternately,

$$\frac{d}{dt} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}, \quad \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$$

or still alternately,

$$\dot{\vec{u}} = A \vec{u}, \quad \vec{x}(0) = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$$

To solve it we assume a solution of the form

$$\begin{aligned} u &= e^{\lambda t} x & \text{so that} & & \dot{u} &= \lambda e^{\lambda t} x \\ v &= e^{\lambda t} y & & & \dot{v} &= \lambda e^{\lambda t} y \end{aligned}$$

plugging this back in we find

$$\lambda e^{\lambda t} x = a e^{\lambda t} x + b e^{\lambda t} y$$

$$\lambda e^{\lambda t} y = c e^{\lambda t} x + d e^{\lambda t} y$$

or

$$\begin{pmatrix} \lambda x \\ \lambda y \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\text{or } A \vec{u} = \lambda \vec{u}$$

Another approach still to this problem is to consider the subspaces or directions a linear transform might find invariant. In other words, we wish to find all vectors $\vec{v}$ s.t.

$$T(\vec{v}) = \lambda \vec{v}, \text{ where } \lambda \in \mathbb{C}.$$

Many other problems in theoretical or practical applications find themselves in this position, that of finding λ s.t. $A\vec{x} = \lambda \vec{x}$.

Definition let $T: V \rightarrow V$ be a linear transform from vector space V into vector space V . A scalar λ is an eigenvalue of T if there is a nonzero vector $\vec{v} \in V$ such that

$$T(\vec{v}) = \lambda \vec{v}.$$

Such a nonzero vector $\vec{v}$ is called an eigenvector of T corresponding to λ .

If the linear transformation T is represented by a $n \times n$ matrix A , where $V = \mathbb{R}^n$, and $T(\vec{v}) = A\vec{v}$, then λ and $\vec{v}$ are characterized by the equation

$$A\vec{v} = \lambda \vec{v}.$$

As we will see later, eigenvalues and eigenvectors represent key physical and mathematical characteristics of the system in consideration.

Notice that $A\vec{v} = \lambda \vec{v} = \lambda I \vec{v}$, so that

$$(A - \lambda I)\vec{v} = \vec{0} \quad \textcircled{*} !$$

Key #1: An eigenvalue is a scalar λ st. the operator $(A - \lambda I)$ is singular
Key #2: An eigenvector is a vector that lives in the nullspace of $(A - \lambda I)$.

According to the Invertible Matrix Theorem, in order to not get the trivial vector $\vec{v} = \vec{0}$, the matrix, or operator, $A - \lambda I$ must be singular.

Also according to the Invertible Matrix Theorem, the determinant of the matrix $A - \lambda I$ must vanish! Thus, an eigenvalue is a number λ such that

$$p(\lambda) = |A - \lambda I| = 0,$$

where $p(\lambda)$ is a polynomial of n^{th} degree called the characteristic polynomial.

Some people treat this as the definition of an eigenvalue, FYI. We will not.

Note that by the Fundamental Theorem of Algebra, $p(\lambda)$ has at most n real roots (and exactly n complex roots). Thus, an $n \times n$ system may have up to n different eigenvalues and n corresponding eigenvectors.

A Demonstrative Example

Ex// $A = \begin{pmatrix} 4 & -5 \\ 2 & -3 \end{pmatrix}$

We are looking for λ such that $|A - \lambda I| = 0$, or

$$\left| \begin{pmatrix} 4 & -5 \\ 2 & -3 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \right| = \left| \begin{pmatrix} 4-\lambda & -5 \\ 2 & -3-\lambda \end{pmatrix} \right| = 0$$

$$\begin{aligned}
 \Rightarrow p(\lambda) &= (4-\lambda)(-3-\lambda) - (-5)(2) = 0 \\
 &= -12 - \lambda + \lambda^2 + 10 = 0 \\
 \lambda^2 - \lambda - 2 &= 0 \\
 (\lambda - 2)(\lambda + 1) &= 0
 \end{aligned}$$

So the matrix has two eigenvalues, $\{-1, 2\}$.

Since the matrix $A - \lambda I$ has a non-trivial nullspace for $\lambda = -1$ and $\lambda = 2$, we set out to find it in the usual way.

$$\lambda = -1 \quad \begin{pmatrix} 4 - (-1) & -5 \\ 2 & -3 - (-1) \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\text{or} \quad \left(\begin{array}{cc|c} 5 & -5 & 0 \\ 2 & -2 & 0 \end{array} \right) \rightarrow \text{RREF} \quad \left(\begin{array}{cc|c} 1 & -1 & 0 \\ 0 & 0 & 0 \end{array} \right)$$

$\therefore v_1 = v_2$, thus the eigenvector, the vector that spans the nullspace of $A - (-1)I$, is

$$\vec{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} v_2 \\ v_2 \end{pmatrix} = v_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Any scalar multiple of $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ works since we have a linear system. We only care about the direction. Technically the unit vector would be ideal, but unnecessary.

$$\text{Again, } \lambda = 2 \quad \begin{pmatrix} 4 - (2) & -5 \\ 2 & -3 - (2) \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\text{or} \quad \left(\begin{array}{cc|c} 2 & -5 & 0 \\ 2 & -5 & 0 \end{array} \right) \rightarrow \text{REF} \quad \left(\begin{array}{cc|c} 2 & -5 & 0 \\ 0 & 0 & 0 \end{array} \right) \quad \begin{array}{l} \text{so } 2v_1 = 5v_2 \\ \text{or } v_1 = \frac{5}{2}v_2 \end{array}$$

$$\text{Thus } \vec{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} \frac{5}{2} v_2 \\ v_2 \end{pmatrix} = v_2 \begin{pmatrix} \frac{5}{2} \\ 1 \end{pmatrix}$$

For convenience, pick $v_2 = 2$, so that the eigenvector corresponding to $\lambda = 2$ is $\vec{v} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$

Summary: If $\lambda = -1$, $\vec{v}_{(-1)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ so that

$$\begin{pmatrix} 4 & -5 \\ 2 & -3 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4-5 \\ 2-3 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix} = -1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \checkmark$$

$$\begin{pmatrix} 4 & -5 \\ 2 & -3 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = (-1) \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

↑
eigenvalue
↑
eigenvector

and if $\lambda = 2$, $\vec{v}_{(2)} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$ so that

$$\begin{pmatrix} 4 & -5 \\ 2 & -3 \end{pmatrix} \begin{pmatrix} 5 \\ 2 \end{pmatrix} = \begin{pmatrix} 20-10 \\ 10-6 \end{pmatrix} = \begin{pmatrix} 10 \\ 4 \end{pmatrix} = 2 \begin{pmatrix} 5 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} 4 & -5 \\ 2 & -3 \end{pmatrix} \begin{pmatrix} 5 \\ 2 \end{pmatrix} = (2) \begin{pmatrix} 5 \\ 2 \end{pmatrix}$$

↑
eigenvalue
↑
eigenvector

///
The situation is the same for 3×3 , 4×4 , etc...

Consider the 2×2 case. $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Note that the characteristic polynomial is given by

$$p(\lambda) = \begin{vmatrix} a-\lambda & b \\ c & d-\lambda \end{vmatrix} = (a-\lambda)(d-\lambda) - bc$$

$$= ad - (a+d)\lambda + \lambda^2 - bc$$

$$= \lambda^2 - (a+d)\lambda + (ad-bc)$$

or $p(\lambda) = \lambda^2 - \text{Tr}(A)\lambda + \text{Det}(A)$ ✓✓

This result is general! Given a characteristic polynomial $p(\lambda) = \lambda^n + a_1\lambda^{n-1} + \dots + a_{n-1}\lambda + a_n$

that factors into linear factors: $p(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2) \dots (\lambda - \lambda_n)$, λ_i being eigenvalues then

"Viète's Formulas"

$$\left\{ \begin{array}{l} \lambda_1 + \lambda_2 + \dots + \lambda_n = \text{Tr}(A) = -a_1 \quad \text{coeff. of } \lambda^{n-1} \text{ (summed one at a time)} \\ \lambda_1\lambda_2 + \lambda_1\lambda_3 + \dots + \lambda_{n-1}\lambda_n = a_2 \quad \text{eigenvalues summed two at a time.} \\ \vdots \\ (\lambda_1 \dots \lambda_k) + (\lambda_1 \dots \lambda_{k-1}\lambda_{k+1}) + \dots + (\lambda_{n-k} \dots \lambda_n) = (-1)^k a_k \quad \text{eigenvalues summed } k \text{ at a time} \\ \vdots \\ (\lambda_1\lambda_2 \dots \lambda_n) = (-1)^n a_n = \det(A) \quad \text{product of all eigenvalues, constant term} \end{array} \right.$$

Viète's Formulas have a great many number of uses! Look up, also, Symmetric Polynomials.

For you: Prove that the determinant is equal to the product of the eigenvalues of a matrix.

For you: What does it mean for a matrix if it has a zero eigenvalue?

Since we made a matrix w/ a non-trivial subspace, $A - \lambda I$, that means each eigenvector spans a subspace of V called the eigenspace of that eigenvalue.

Defn:

For each eigenvalue λ of a linear transformation $T: V \rightarrow V$, the eigenspace

$$V_\lambda = E_\lambda = \{ \vec{v} \in V \mid T(\vec{v}) = \lambda \vec{v} \}$$

$\uparrow$ everyone else $\uparrow$ book

is a subspace of V .

Ex/ Given $\begin{pmatrix} 4 & -5 \\ 2 & -3 \end{pmatrix}$, the eigenvalues were $\{-1, 2\}$. Thus,

$$\mathbb{V}_{(-1)} = \text{span} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{and} \quad \mathbb{V}_{(2)} = \text{span} \begin{pmatrix} 5 \\ 2 \end{pmatrix}$$

Obvious question: Can two eigenvectors of different eigenvalues be dependent?

Theorem: Let A be an $n \times n$ matrix. If $\lambda_1, \lambda_2, \dots, \lambda_n$ are distinct eigenvalues w/ corresponding eigenvectors $\vec{v}_1, \dots, \vec{v}_n$, then $\{\vec{v}_1, \dots, \vec{v}_n\}$ are linearly independent.

Pf: By induction and contradiction. Obviously one vector is independent.

Suppose k eigenvectors are linearly independent. I.e.

$$c_1 \vec{v}_1 + \dots + c_k \vec{v}_k = \vec{0} \quad \text{only if} \quad c_1 = \dots = c_k = 0$$

Then we need to prove $k+1$ vectors are independent. In other words, we suppose

$$c_1 \vec{v}_1 + \dots + c_k \vec{v}_k + c_{k+1} \vec{v}_{k+1} = \vec{0} \quad (*) \quad \text{w/ "proving" dependency.}$$

Operate A on the system. Obviously $A\vec{0} = \vec{0}$ so

$$c_1 \lambda_1 \vec{v}_1 + \dots + c_k \lambda_k \vec{v}_k + c_{k+1} \lambda_{k+1} \vec{v}_{k+1} = \vec{0}$$

Take (*) and multiply by λ_{k+1} and subtract from the above to get

$$c_1 (\lambda_1 - \lambda_{k+1}) \vec{v}_1 + \dots + c_k (\lambda_k - \lambda_{k+1}) \vec{v}_k + \underbrace{c_{k+1} (\lambda_{k+1} - \lambda_{k+1})}_{=0} \vec{v}_{k+1} = \vec{0}$$

or,

$$c_1(\lambda_1 - \lambda_{k+1})\vec{v}_1 + \dots + c_k(\lambda_k - \lambda_{k+1})\vec{v}_k = \vec{0}$$

Since we assumed dependency, $c_i(\lambda_i - \lambda_{k+1}) = 0$ for $c_i \neq 0$ for some i .

But this violates the hypothesis that all eigenvalues are distinct, so dependency is false. The theorem is proved.

Another question: If eigenvalues are roots of a characteristic polynomial, then there is a chance for repeated eigenvalues. How does one handle that case?

Answer: Exactly the same way! You're still looking for the nullspace of $(A - \lambda I)$!

Ex// $A = \begin{pmatrix} 5 & -6 & -6 \\ -1 & 4 & 2 \\ 3 & -6 & -4 \end{pmatrix}$. The characteristic polynomial is given by

$$p(\lambda) = \begin{vmatrix} 5-\lambda & -6 & -6 \\ -1 & 4-\lambda & 2 \\ 3 & -6 & -4-\lambda \end{vmatrix} \xrightarrow{-C_3 + C_2} \begin{vmatrix} 5-\lambda & 0 & -6 \\ -1 & 2-\lambda & 2 \\ 3 & \lambda-2 & -4-\lambda \end{vmatrix} = (\lambda-2) \begin{vmatrix} 5-\lambda & 0 & -6 \\ -1 & -1 & 2 \\ 3 & 1 & -4-\lambda \end{vmatrix}$$

$$R_3 + R_2 \quad (\lambda-2) \begin{vmatrix} 5-\lambda & 0 & -6 \\ 2 & 0 & -2-\lambda \\ 3 & 1 & -4-\lambda \end{vmatrix} = -(\lambda-2) \begin{vmatrix} 5-\lambda & -6 \\ 2 & -2-\lambda \end{vmatrix} = (\lambda-2) \begin{vmatrix} 5-\lambda & 6 \\ 2 & 2+\lambda \end{vmatrix}$$

$$= (\lambda-2) [(5-\lambda)(2+\lambda) - 12] = (\lambda-2) [10 + 3\lambda - \lambda^2 - 12] = -(\lambda-2) [\lambda^2 - 3\lambda + 2] \\ = -(\lambda-2)(\lambda-1)(\lambda-2) = -(\lambda-1)(\lambda-2)^2$$

Eigenvalues are $\{1, 2\}$

$$A - (1)I = \begin{pmatrix} 4 & -6 & -6 \\ -1 & 3 & 2 \\ 3 & -6 & -5 \end{pmatrix} \quad \text{and}$$

$$A - (2)I = \begin{pmatrix} 3 & -6 & -6 \\ -1 & 2 & 2 \\ 3 & -6 & -6 \end{pmatrix}$$

Notice that both of these matrices must be singular (we made them that way!). What are their nullspaces, and thus the eigenvectors?

$$\begin{aligned} \lambda=1 & \quad \left(\begin{array}{ccc|c} 4 & -6 & -6 & 0 \\ -1 & 3 & 2 & 0 \\ 3 & -6 & -5 & 0 \end{array} \right) \xrightarrow[R_2 \leftrightarrow R_1]{-R_1} \left(\begin{array}{ccc|c} 1 & -3 & -2 & 0 \\ 4 & -6 & -6 & 0 \\ 3 & -6 & -5 & 0 \end{array} \right) \xrightarrow[-3R_1+R_3]{-4R_1+R_2} \left(\begin{array}{ccc|c} 1 & -3 & -2 & 0 \\ 0 & 6 & 2 & 0 \\ 0 & 3 & 1 & 0 \end{array} \right) \\ & \quad -\frac{R_2}{2} + R_3 \rightarrow \left(\begin{array}{ccc|c} 1 & -3 & -2 & 0 \\ 0 & 6 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \quad \text{Rank 2} \end{aligned}$$

Basic Var.: v_1, v_2 Free Var.: v_3

$$3v_2 = -v_3 \Rightarrow v_2 = -\frac{1}{3}v_3$$

$$v_1 - 3v_2 - 2v_3 = 0 \Rightarrow v_1 = 3v_2 + 2v_3 = -v_3 + 2v_3 = v_3$$

$$\therefore \vec{v}_{(1)} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} v_3 \\ -\frac{1}{3}v_3 \\ v_3 \end{pmatrix} = \frac{1}{3}v_3 \begin{pmatrix} 3 \\ -1 \\ 3 \end{pmatrix} \quad \text{The eigenspace corresponding to } \lambda=1 \text{ is one-dimensional}$$

↑ eigenvector corresponding to $\lambda=1$

$$\lambda=2 \quad \left(\begin{array}{ccc|c} 3 & -6 & -6 & 0 \\ -1 & 2 & 2 & 0 \\ 3 & -6 & -6 & 0 \end{array} \right) \xrightarrow[\text{Obs.}]{\text{RREF}} \left(\begin{array}{ccc|c} 1 & -2 & -2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \quad \begin{array}{l} \text{Basic Var.: } v_1 \\ \text{Free var.: } v_2 \neq v_3 \end{array}$$

$$v_1 = 2v_2 + 2v_3 \quad \therefore \vec{v}_{(2)} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 2v_2 + 2v_3 \\ v_2 \\ v_3 \end{pmatrix} = v_2 \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + v_3 \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$$

The eigenspace corresponding to $\lambda=2$ is two-dimensional. Both $\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$ & $\begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$ are eigenvectors.

$$A\vec{v} = \lambda\vec{v}$$

Check!

$$A\vec{v}_{(1)} = \begin{pmatrix} 5 & -6 & -6 \\ -1 & 4 & 2 \\ 3 & -6 & -4 \end{pmatrix} \begin{pmatrix} 3 \\ -1 \\ 3 \end{pmatrix} = \begin{pmatrix} 15 + 6 - 18 \\ -3 - 4 + 6 \\ 9 + 6 - 12 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \\ 3 \end{pmatrix} \quad \checkmark$$

$$A\vec{v}_{(2),2} = \begin{pmatrix} 5 & -6 & -6 \\ -1 & 4 & 2 \\ 3 & -6 & -4 \end{pmatrix} \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 10 - 6 \\ -2 + 2 \\ 6 - 4 \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix} = 2 \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} \quad \checkmark$$

$$A\vec{v}_{(2),1} = \begin{pmatrix} 5 & -6 & -6 \\ -1 & 4 & 2 \\ 3 & -6 & -4 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 10 - 6 \\ -2 + 4 \\ 6 - 6 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ 0 \end{pmatrix} = 2 \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \quad \checkmark$$

Objective: You are looking for the Nullspace of $(A - \lambda I)$ for all eigenvalues.
Same λ same λ